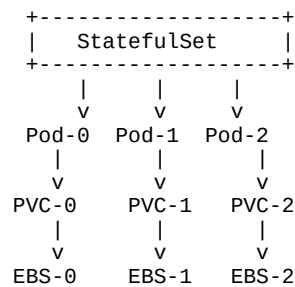


StatefulSet + AWS EBS Architecture Guide

1. What is a StatefulSet?

A StatefulSet is used to manage stateful applications like databases (MySQL, PostgreSQL, MongoDB). It provides stable network identity, stable storage, and ordered deployment/scaling.

2. Architecture Overview



Each replica gets its own dedicated EBS volume.

3. How Dynamic Provisioning Works

1. StatefulSet creates Pod-0
2. volumeClaimTemplates create PVC-0
3. StorageClass triggers EBS CSI Driver
4. AWS creates EBS volume
5. Volume attaches to worker node
6. Pod mounts the volume

This process repeats for each replica.

4. Example YAML Configuration

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
  name: mysql
spec:
  serviceName: mysql
  replicas: 2
  selector:
    matchLabels:
      app: mysql
  template:
    metadata:
      labels:
        app: mysql
    spec:
      containers:
```

```
- name: mysql
  image: mysql:8
  volumeMounts:
    - name: mysql-data
      mountPath: /var/lib/mysql
volumeClaimTemplates:
- metadata:
  name: mysql-data
  spec:
    accessModes: ["ReadWriteOnce"]
    storageClassName: ebs-sc
    resources:
      requests:
        storage: 20Gi
```

5. Multi-AZ Consideration

EBS volumes are AZ-specific.

If Pod-0 runs in AZ-1a → EBS-0 created in AZ-1a

If node fails and reschedules to AZ-1b → volume cannot attach.

Solution:

- Use node affinity
- Use WaitForFirstConsumer
- Or use EFS for shared storage